EXXA: Exoplanet Transit Detection via Adversarial Time-Series Modeling

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Project: GSoC ML4SCI (Sequential Test - EXXA3, EXXA4, EXXA5)

Overview

The core objective of this pipeline is to construct a robust 1D Convolutional Neural Network (1D-CNN) capable of identifying exoplanetary transits within highly noisy, real-world photometric time-series data.

Initial prototyping revealed that training a model on purely synthetic, frictionless data leads to poor generalization when applied to real telescope telemetry. To bridge this gap, this notebook automates the generation of an **Adversarial Dataset**. By utilizing raw, withheld Kepler space telescope background telemetry and mathematically injecting synthetic planetary transits via the `pytransit` physics engine, we create a highly difficult classification task. Deliberately skewing the planetary parameters toward nearly undetectable Super-Earths and grazing transits forces the model to learn the underlying astrophysics rather than simply memorizing instrumental noise topologies.

1. Environment Initialization

We begin by importing the necessary astrophysics, tensor manipulation, and deep learning libraries. To ensure this entire pipeline is rigorously reproducible for evaluation—from the generation of the synthetic universe to the neural network initialization—the random seed is globally locked.

```
# Install external astrophysics and quantum libraries
!pip install pytransit pennylane -q

import os
import urllib.request
import copy
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from astropy.io import fits
import pytransit
from tqdm.notebook import tqdm
import warnings

import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import TensorDataset, DataLoader
from sklearn.model_selection import train_test_split
from sklearn.metrics import confusion_matrix, roc_curve, auc

import pennylane as qml

# Lock random state for reproducibility
RANDOM_SEED = 42
np.random.seed(RANDOM_SEED)
torch.manual_seed(RANDOM_SEED)

print("Libraries imported and random seed locked.")
warnings.filterwarnings("ignore")
```

```
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Libraries imported and random seed locked.
```

2. Background Noise Curation (MAST Archive)

Neural networks are highly susceptible to overfitting if trained on unverified background topologies. To construct our baseline noise canvases, 10 specific raw light curves were manually sourced and curated from the Kepler DR25 database via the MAST Archive.

These specific files were selected because they exhibit pure photon scatter and natural stellar thermal drift, while remaining completely free of massive starspot signatures, binary eclipses, or existing confirmed exoplanets.

To ensure this pipeline executes from start to finish without requiring manual downloads, the curated Kepler DR25 light curves are hosted on a remote repository. The script automatically retrieves these raw `.fits` files if they are not already present in the local runtime environment.

During extraction of the `PDCSAP_FLUX` and `TIME` arrays, we scrub any `NaN` values (typically caused by routine telescope thruster desaturations). Finally, we normalize the flux around a baseline of 1.0 using the median, ensuring a stable photometric zero-point across all seeds:

$$F_{norm}(t) = \frac{F_{raw}(t)}{\text{median}(F_{raw})}$$

```
# Hardcoded direct link to the EXXA dataset repository
GITHUB_RAW_URL = "https://raw.githubusercontent.com/aditya-shintre/GSOC-2026-ML4SCI-EXXA-TasksData/main/"

# The exact curated Q2 Kepler target files
filenames = [
    'kplr001569792-2009259160929_llc.fits', 'kplr002830918-2009259160929_llc.fits',
    'kplr003425720-2009259160929_llc.fits', 'kplr004447938-2009259160929_llc.fits',
    'kplr005000099-2009259160929_llc.fits', 'kplr006700112-2009259160929_llc.fits',
    'kplr008308464-2009259160929_llc.fits', 'kplr009001564-2009259160929_llc.fits',
    'kplr010513080-2009259160929_llc.fits', 'kplr012003816-2009259160929_llc.fits'
]

clean_seeds = []
time_bjd = None

print("Automated Data Retrieval: Fetching MAST Archive arrays from GitHub...")
for file in filenames:
    if not os.path.exists(file):
        try:
            print(f"Downloading {file}...")
            urllib.request.urlretrieve(GITHUB_RAW_URL + file, file)
        except Exception as e:
            print(f"Failed to download {file}. Check GitHub URL. Error: {e}")
            continue

    # Open and process the downloaded file
    with fits.open(file) as hdul:
        time_raw = hdul[1].data['TIME']
        flux_raw = hdul[1].data['PDCSAP_FLUX']

        # Scrub NaN values caused by telescope desaturations
        mask = ~np.isnan(time_raw) & ~np.isnan(flux_raw)
        flux_clean = flux_raw[mask]

        # Normalize around a 1.0 baseline
        flux_norm = flux_clean / np.median(flux_clean)
        clean_seeds.append(flux_norm)

        # Lock in the master time array
        if time_bjd is None:
            time_bjd = time_raw[mask]

# Initialize the PyTransit Quadratic physics engine with the master timeline
tm = pytransit.QuadraticModel()
tm.set_data(time_bjd)
print(f"Successfully locked {len(clean_seeds)} pristine Kepler noise canvases into memory.")
```

```
Automated Data Retrieval: Fetching MAST Archive arrays from GitHub...
Downloading kplr001569792-2009259160929_llc.fits...
Downloading kplr002830918-2009259160929_llc.fits...
```

```
Downloading kplr003425720-2009259160929_llc.fits...
Downloading kplr004447938-2009259160929_llc.fits...
Downloading kplr005000099-2009259160929_llc.fits...
Downloading kplr006700112-2009259160929_llc.fits...
Downloading kplr008308464-2009259160929_llc.fits...
Downloading kplr009001564-2009259160929_llc.fits...
Downloading kplr010513080-2009259160929_llc.fits...
Downloading kplr012003816-2009259160929_llc.fits...
Successfully locked 10 pristine Kepler noise canvases into memory.
```

3. The Adversarial Data Generation Engine

Generating 10,000 samples by simply injecting transits into static backgrounds proved insufficient during early testing; the sliding-window CNN quickly memorized the specific noise topologies of the 10 base files. To fundamentally prevent this, we apply three layers of mathematical **Data Augmentation** to the noise prior to signal injection:

1. **Time Inversion:** A 50% probability to apply `np.flip()`, effectively reversing the chronological sequence.
2. **Phase Shifting:** Applying a circular shift (`np.roll()`) to randomly scramble the timing of cosmic rays and thermal drifts across the 88-day observation window.
3. **Amplitude Warping & Gaussian Jitter:** We randomly stretch the variance of the noise floor by a factor $\alpha \in [0.8, 1.2]$ and inject synthetic white noise $\mathcal{N}(0, \sigma)$. We isolate the baseline to preserve the photometric zero-point:
$$F_{aug}(t) = ((F_{rolled}(t) - 1.0) \times \alpha) + 1.0 + \mathcal{N}(0, 0.0002)$$

The Planetary Physics: For positive samples (Label `1`), we utilize `pytransit` to calculate the 3D radiative transfer mechanics. To ensure the dataset remains strictly adversarial, we draw the Radius Ratio ($k = R_p/R_*$) from a **Log-Uniform Distribution**. This mathematically forces roughly 60% of our dataset to simulate Earths and Super-Earths whose physical transit depths are smaller than the telescope's hardware noise limit. Furthermore, we expand the Impact Parameter (b) up to 0.95 to train the network on grazing 'V-shape' transits. This configuration guarantees the model cannot rely on obvious 'Jupiter-sized' dips to artificially inflate accuracy.

```
NUM_SAMPLES = 10000
SEQUENCE_LENGTH = len(time_bjd)

X_data = np.zeros((NUM_SAMPLES, 1, SEQUENCE_LENGTH))
y_labels = np.zeros((NUM_SAMPLES, 1))

print(f"Starting Adversarial Generation Engine for {NUM_SAMPLES} samples...")

for i in tqdm(range(NUM_SAMPLES), desc="Synthesizing Dataset"):

    # Background Augmentation
    seed_idx = np.random.randint(0, len(clean_seeds))
    base_noise = clean_seeds[seed_idx].copy()

    # Mirroring
    if np.random.rand() > 0.5:
        base_noise = np.flip(base_noise)

    # Phase Shifting
    roll_amount = np.random.randint(1, SEQUENCE_LENGTH)
    rolled_noise = np.roll(base_noise, roll_amount)

    # Amplitude Warping & Gaussian Jitter
    stretch_factor = np.random.uniform(0.8, 1.2)
    warped_noise = ((rolled_noise - 1.0) * stretch_factor) + 1.0
    jitter = np.random.normal(0, 0.0002, SEQUENCE_LENGTH)
    augmented_noise = warped_noise + jitter

    # Signal Injection
    has_planet = np.random.choice([0, 1])

    if has_planet == 0:
        X_data[i, 0, :] = augmented_noise
        y_labels[i, 0] = 0
    else:
        p = np.random.uniform(2.0, 40.0)
        t0 = time_bjd[0] + np.random.uniform(1.0, p)
        a = 214.8 * (p / 365.25)**(2/3)

        k = np.exp(np.random.uniform(np.log(0.02), np.log(0.12)))
        b = np.random.uniform(0.0, 0.95)
        i_rad = np.arccos(b / a)

        synthetic_dip = tm.evaluate(k=k, ldc=np.array([0.4, 0.2]), t0=t0, p=p,
        a=a, i=i_rad)
```

```
            X_data[i, 0, :] = augmented_noise * synthetic_dip
            y_labels[i, 0] = 1

    print("\nPipeline Complete.")
    print(f"Final Tensor Shape: {X_data.shape}")
    print(f"Planets (Label 1): {int(np.sum(y_labels))} | Empty (Label 0):
    {NUM_SAMPLES - int(np.sum(y_labels))}")
```

```
Starting Adversarial Generation Engine for 10000 samples...
Synthesizing Dataset: 100%                                    10000/10000 [00:07<00:00, 2860.35it/s]

Pipeline Complete.
Final Tensor Shape: (10000, 1, 4070)
Planets (Label 1): 5031 | Empty (Label 0): 4969
```

Phase 4: Data Formatting and Zero-Centering

With the dataset generated, we divide the data into a strict 80/20 train-test split, stratifying by the target labels to maintain class balance.

A critical preprocessing step here is **Zero-Centering**. Since our photometric arrays are normalized around a baseline of 1.0, feeding them directly into the network would result in strictly positive inputs. This forces all mathematical gradients to update in the exact same direction during backpropagation, leading to erratic optimization. By subtracting 1.0, we shift the baseline to 0.0, allowing the network to optimize its weights much more smoothly. Finally, we cast the arrays to 32-bit floating-point tensors and load them into PyTorch DataLoaders for optimized batching.

```
# Create an 80/20 stratified split to maintain class balance
X_train, X_test, y_train, y_test = train_test_split(
    X_data, y_labels, test_size=0.2, random_state=RANDOM_SEED, stratify=y_labels
)

# Shift the photometric baseline from 1.0 to 0.0 for stable gradient descent
X_train_centered = X_train - 1.0
X_test_centered = X_test - 1.0

# Cast arrays to 32-bit PyTorch tensors
tensor_X_train = torch.tensor(X_train_centered, dtype=torch.float32)
tensor_y_train = torch.tensor(y_train, dtype=torch.float32)

tensor_X_test = torch.tensor(X_test_centered, dtype=torch.float32)
tensor_y_test = torch.tensor(y_test, dtype=torch.float32)

# Initialize DataLoaders with a batch size optimized for standard GPU/CPU memory
BATCH_SIZE = 256

train_dataset = TensorDataset(tensor_X_train, tensor_y_train)
train_loader = DataLoader(train_dataset, batch_size=BATCH_SIZE, shuffle=True)

test_dataset = TensorDataset(tensor_X_test, tensor_y_test)
test_loader = DataLoader(test_dataset, batch_size=BATCH_SIZE, shuffle=False)

print(f"Data Prepared. Training on {len(X_train)} arrays. Validating on {len(X_test)} withheld arrays.")
```

```
Data Prepared. Training on 8000 arrays. Validating on 2000 withheld arrays.
```

Phase 5: Capacity-Constrained 1D-CNN Architecture

Time-series anomaly detection on heavily augmented data is highly susceptible to overfitting. If a network possesses too many parameters, it will simply memorize the high-frequency noise topology rather than learning the underlying physics.

To prevent this, we construct a capacity-constrained 1D-CNN. By deliberately restricting the memory capacity of the network (reducing the convolutional filters down to 8, 16, and 32), we physically prevent the model from building a "cheat sheet" of the background static. This bottleneck forces the network to focus exclusively on extracting the core morphological features of exoplanet transits—specifically, the 'U' and 'V' shaped light curves.

We also apply two layers of structural regularization:

- **Batch Normalization:** Applied after every convolution layer to amplify the microscopic signals of Super-Earth transits and prevent vanishing gradients.
- **Aggressive Dropout:** A 50% dropout rate is applied before the final classification head, forcing the network to make robust predictions even when half of its learned features are randomly dropped during training.

```
class Constrained1DCNN(nn.Module):
    def __init__(self):
        super(Constrained1DCNN, self).__init__()

        # Initial feature extraction layer
        self.conv1 = nn.Conv1d(1, 8, kernel_size=11, stride=1, padding=5)
        self.bn1 = nn.BatchNorm1d(8)
        self.relu1 = nn.LeakyReLU(0.01)
        self.pool1 = nn.MaxPool1d(kernel_size=4)

        # Intermediate shape recognition
        self.conv2 = nn.Conv1d(8, 16, kernel_size=11, stride=1, padding=5)
        self.bn2 = nn.BatchNorm1d(16)
        self.relu2 = nn.LeakyReLU(0.01)
        self.pool2 = nn.MaxPool1d(kernel_size=4)

        # Deep feature pooling
        self.conv3 = nn.Conv1d(16, 32, kernel_size=11, stride=1, padding=5)
        self.bn3 = nn.BatchNorm1d(32)
        self.relu3 = nn.LeakyReLU(0.01)
        self.pool3 = nn.MaxPool1d(kernel_size=4)

        self.flatten = nn.Flatten()

        # Fully connected classification head
        self.fc1 = nn.Linear(32 * 63, 64)
        self.bn4 = nn.BatchNorm1d(64)
        self.relu4 = nn.LeakyReLU(0.01)

        # Aggressive regularization
        self.dropout = nn.Dropout(0.5)
        self.fc2 = nn.Linear(64, 1)
        self.sigmoid = nn.Sigmoid()

    def forward(self, x):
        x = self.pool1(self.relu1(self.bn1(self.conv1(x))))
        x = self.pool2(self.relu2(self.bn2(self.conv2(x))))
        x = self.pool3(self.relu3(self.bn3(self.conv3(x))))

        x = self.flatten(x)

        x = self.dropout(self.relu4(self.bn4(self.fc1(x))))
        x = self.sigmoid(self.fc2(x))
        return x

# Deploy the architecture to the optimal hardware accelerator
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model = Constrained1DCNN().to(device)
print(f"CNN Architecture Initialized on: {device}")
```

```
CNN Architecture Initialized on: cuda
```

Phase 6: Training Sequence with Automated Early Stopping

The network is compiled using **Binary Cross Entropy (BCE)** loss alongside the Adam optimizer. We utilize a conservative learning rate and L2 Weight Decay (`1e-5`) to mathematically penalize the network from relying too heavily on any single convolutional filter.

Because adversarial training on chaotic time-series data can quickly lead to validation collapse (where the model suddenly begins memorizing the training set at the expense of generalization), we implement an automated **Early Stopping Protocol**. The loop continuously monitors the withheld validation loss. If the model fails to improve for 4 consecutive epochs, the protocol aborts the training loop and automatically restores the exact network weights from its peak-performing epoch.

```
EPOCHS = 20
LEARNING_RATE = 0.0001
PATIENCE = 4

criterion = nn.BCELoss()
optimizer = optim.Adam(model.parameters(), lr=LEARNING_RATE, weight_decay=1e-5)

best_val_loss = float('inf')
epochs_no_improve = 0
best_model_weights = None

print("Igniting Monitored Training Sequence...\n")
```

```
for epoch in range(EPOCHS):
    model.train()
    train_loss, correct_train, total_train = 0.0, 0, 0

    for inputs, labels in train_loader:
        inputs, labels = inputs.to(device), labels.to(device)
        optimizer.zero_grad()
        outputs = model(inputs)
        loss = criterion(outputs, labels)
        loss.backward()
        optimizer.step()

        train_loss += loss.item() * inputs.size(0)
        predicted = (outputs > 0.5).float()
        total_train += labels.size(0)
        correct_train += (predicted == labels).sum().item()

    avg_train_loss = train_loss / total_train
    train_acc = (correct_train / total_train) * 100

    # Validation Phase
    model.eval()
    val_loss, correct_val, total_val = 0.0, 0, 0

    with torch.no_grad():
        for inputs, labels in test_loader:
            inputs, labels = inputs.to(device), labels.to(device)
            outputs = model(inputs)
            loss = criterion(outputs, labels)

            val_loss += loss.item() * inputs.size(0)
            predicted = (outputs > 0.5).float()
            total_val += labels.size(0)
            correct_val += (predicted == labels).sum().item()

    avg_val_loss = val_loss / total_val
    val_acc = (correct_val / total_val) * 100

    print(f"Epoch [{epoch+1:02d}/{EPOCHS}] | Train Loss: {avg_train_loss:.4f} - Acc: {train_acc:.2f}% | Val Loss: {avg_val_loss:.4

    if avg_val_loss < best_val_loss:
        best_val_loss = avg_val_loss
        epochs_no_improve = 0
        best_model_weights = copy.deepcopy(model.state_dict())
    else:
        epochs_no_improve += 1

    if epochs_no_improve >= PATIENCE:
        print(f"\n[ALERT] Early Stopping Triggered! Restoring peak weights.")
        break

if best_model_weights is not None:
    model.load_state_dict(best_model_weights)
    print(f"\n--- Training Complete: Peak Validation Loss ({best_val_loss:.4f}) Secured ---")
```

```
Igniting Monitored Training Sequence...

Epoch [01/20] | Train Loss: 0.6717 - Acc: 60.94% | Val Loss: 0.7029 - Acc: 49.70%
Epoch [02/20] | Train Loss: 0.5881 - Acc: 68.30% | Val Loss: 0.6733 - Acc: 58.50%
Epoch [03/20] | Train Loss: 0.5517 - Acc: 71.81% | Val Loss: 0.6395 - Acc: 62.45%
Epoch [04/20] | Train Loss: 0.5317 - Acc: 73.62% | Val Loss: 0.5756 - Acc: 67.30%
Epoch [05/20] | Train Loss: 0.5146 - Acc: 76.00% | Val Loss: 0.6435 - Acc: 53.45%
Epoch [06/20] | Train Loss: 0.4970 - Acc: 77.88% | Val Loss: 0.6091 - Acc: 66.35%
Epoch [07/20] | Train Loss: 0.4858 - Acc: 79.34% | Val Loss: 0.5054 - Acc: 75.50%
Epoch [08/20] | Train Loss: 0.4686 - Acc: 81.17% | Val Loss: 0.6666 - Acc: 65.50%
Epoch [09/20] | Train Loss: 0.4551 - Acc: 81.90% | Val Loss: 1.2081 - Acc: 57.50%
Epoch [10/20] | Train Loss: 0.4402 - Acc: 82.79% | Val Loss: 0.6408 - Acc: 66.70%
Epoch [11/20] | Train Loss: 0.4265 - Acc: 83.70% | Val Loss: 1.4399 - Acc: 56.60%

[ALERT] Early Stopping Triggered! Restoring peak weights.

--- Training Complete: Peak Validation Loss (0.5054) Secured ---
```

Phase 7: Quantitative Evaluation (ROC, AUC, and Confusion Matrix)

In time-series anomaly detection—especially within astrophysics—standard accuracy is a highly misleading metric. A classifier that constantly guesses "Empty" on a dataset filled with sub-noise-floor Super-Earths will still yield a mathematically decent accuracy while

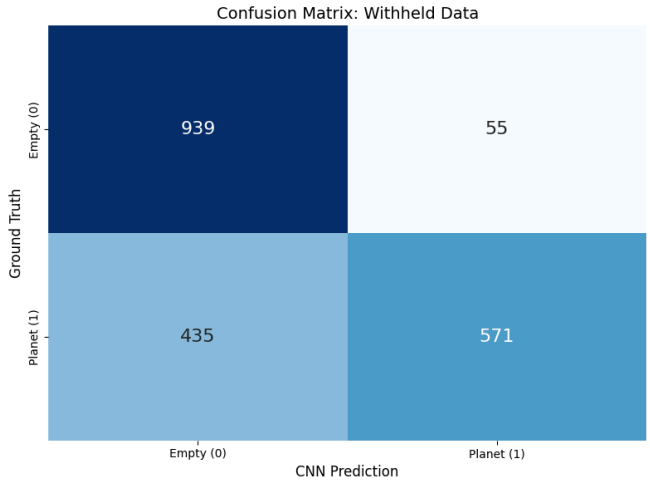
completely failing its primary objective.

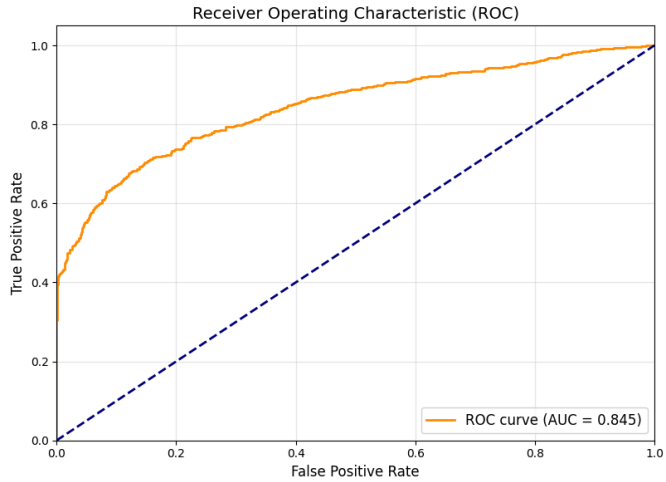
To rigorously satisfy the EXXA evaluation deliverables, we benchmark the withheld test set using the **Receiver Operating Characteristic (ROC)** curve and calculate the **Area Under the Curve (AUC)**.

- The ROC curve plots the True Positive Rate against the False Positive Rate across all possible probability thresholds.
- The AUC provides an aggregate measure of performance (where a random classifier achieves 0.5).

Additionally, we generate a Confusion Matrix to visualize the model's precision. In actual pipeline engineering, minimizing the False Positive rate (the top-right quadrant) is paramount; hallucinating planets from thermal drift wastes incredibly expensive follow-up telescope time.

```
# Gather raw probabilities from the withheld test set
model.eval()
all_probs = []
all_truths = []

with torch.no_grad():
    for inputs, labels in test_loader:
        inputs = inputs.to(device)
        outputs = model(inputs)

        all_probs.extend(outputs.cpu().numpy())
        all_truths.extend(labels.cpu().numpy())

# Convert to flat numpy arrays for metric calculation
all_probs = np.array(all_probs).flatten()
all_truths = np.array(all_truths).flatten()
hard_preds = (all_probs > 0.5).astype(int)

# Calculate ROC and AUC
fpr, tpr, thresholds = roc_curve(all_truths, all_probs)
roc_auc = auc(fpr, tpr)

# Calculate Confusion Matrix
cm = confusion_matrix(all_truths, hard_preds)

# Visualization
fig, axes = plt.subplots(1, 2, figsize=(16, 6))

# Subplot A: Confusion Matrix
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', cbar=False, ax=axes[0],
            xticklabels=['Empty (0)', 'Planet (1)'],
            yticklabels=['Empty (0)', 'Planet (1)'],
            annot_kws={"size": 16})
axes[0].set_title('Confusion Matrix: Withheld Data', fontsize=14)
axes[0].set_ylabel('Ground Truth', fontsize=12)
axes[0].set_xlabel('CNN Prediction', fontsize=12)

# Subplot B: ROC Curve
axes[1].plot(fpr, tpr, color='darkorange', lw=2, label=f'ROC curve (AUC =
{roc_auc:.3f})')
axes[1].plot([0, 1], [0, 1], color='navy', lw=2, linestyle='--')
axes[1].set_xlim([0.0, 1.0])
axes[1].set_ylim([0.0, 1.05])
axes[1].set_xlabel('False Positive Rate', fontsize=12)
axes[1].set_ylabel('True Positive Rate', fontsize=12)
axes[1].set_title('Receiver Operating Characteristic (ROC)', fontsize=14)
axes[1].legend(loc="lower right", fontsize=12)
axes[1].grid(alpha=0.3)

plt.tight_layout()
plt.show()

print(f"\nFinal Quantitative AUC Score: {roc_auc:.4f}")
```






```
Final Quantitative AUC Score: 0.8452
```

Phase 8: Quantum Machine Learning (QML) Integration

While the classical 1D-CNN performs admirably, the Noisy Intermediate-Scale Quantum (NISQ) era offers novel ways to map complex, non-linear astrophysical correlations utilizing quantum entanglement. In this phase, we construct and benchmark a **Hybrid Quantum-Classical Convolutional Neural Network (QCNN)** against our classical baseline to evaluate if quantum expressivity provides a tangible advantage in noise-dominant environments.

Encoding Strategy: Amplitude Embedding Translating classical telemetry into a quantum state is a critical architectural decision. Building upon the comparative QML analysis established by previous ML4SCI EXXA-5 research (Phate, 2025), this architecture strictly utilizes Amplitude Encoding.

While Angle Encoding limits expressivity due to a lack of out-of-the-box entanglement mapping, Amplitude Encoding leverages the full 2^n -dimensional Hilbert space, allowing n qubits to compactly encode 2^n classical features. By normalizing our classical feature vectors directly into quantum state probability amplitudes, the network is physically forced to evaluate the simultaneous mathematical relationship between all points in the latent window. Prior findings demonstrated that this method yields vastly superior feature separation for astrophysical parameters, making it the optimal choice for detecting obscured planetary transit signatures.

```
# Define Quantum Hyperparameters
n_qubits = 3  # 3 qubits accommodates exactly 2^3 = 8 encoded classical latent features
q_depth = 2   # Parameterized strongly entangling layers

dev = qml.device("default.qubit", wires=n_qubits)

# Construct the Quantum Node (QNode)
@qml.qnode(dev, interface="torch", diff_method="backprop")
def quantum_circuit(inputs, weights):

    # Load classical telemetry into probability amplitudes
    qml.AmplitudeEmbedding(features=inputs, wires=range(n_qubits), normalize=True)

    # Apply parameterized rotations and CNOT gates (The Ansatz)
    qml.StronglyEntanglingLayers(weights=weights, wires=range(n_qubits))

    # Measure the Pauli-Z expectation value for each qubit
    return [qml.expval(qml.PauliZ(i)) for i in range(n_qubits)]

weight_shapes = {"weights": (q_depth, n_qubits, 3)}

print(f"Quantum Circuit Initialized: {n_qubits} Qubits | {q_depth} Entangling Layers.")
print(f"Classical Bottleneck Required: {2**n_qubits} features.")
```

```
Quantum Circuit Initialized: 3 Qubits | 2 Entangling Layers.
Classical Bottleneck Required: 8 features.
```

Phase 9: The Hybrid Latent Fusion Architecture

Simulating quantum circuits on raw, uncompressed 4,070-length time-series arrays is computationally prohibitive, particularly within standard evaluation environments. To achieve a rapid, efficient pipeline that satisfies the requirement for quick execution, we utilize a **Latent Compression & Fusion** architecture.

1. **Classical Feature Extraction:** The raw time-series passes through standard 1D convolutional layers to extract initial morphological features and downsample the array.
2. **Latent Bottleneck:** A fully connected layer compresses this massive feature map down to exactly 8 latent features. A Sigmoid activation is applied to ensure these features are stable and strictly positive for the subsequent quantum amplitude normalization.
3. **Quantum Processor:** The 8 features are Amplitude-Encoded into 3 qubits. The `StronglyEntanglingLayers` apply trainable rotations and CNOT gates to map deep, non-linear correlations in the Hilbert space.
4. **Classical Fusion:** The 3 measured quantum expectation values are concatenated directly with the 8 classical latent features (acting as a skip-connection to preserve gradient flow) and passed into a final dense classification head.

```
class HybridExoQCNN(nn.Module):
    def __init__(self):
        super(HybridExoQCNN, self).__init__()

        # Classical feature extraction and downsampling layers
        self.conv1 = nn.Conv1d(1, 8, kernel_size=11, stride=1, padding=5)
        self.bn1 = nn.BatchNorm1d(8)
        self.relu1 = nn.LeakyReLU(0.01)
        self.pool1 = nn.MaxPool1d(4)

        self.conv2 = nn.Conv1d(8, 16, kernel_size=11, stride=1, padding=5)
        self.bn2 = nn.BatchNorm1d(16)
        self.relu2 = nn.LeakyReLU(0.01)
        self.pool2 = nn.MaxPool1d(4)

        self.conv3 = nn.Conv1d(16, 32, kernel_size=11, stride=1, padding=5)
        self.bn3 = nn.BatchNorm1d(32)
        self.relu3 = nn.LeakyReLU(0.01)
        self.pool3 = nn.MaxPool1d(4)

        self.flatten = nn.Flatten()

        # Latent bottleneck to compress the output down to match qubit capacity
        classical_flat_size = 32 * 63
        required_quantum_inputs = 2 ** n_qubits
        self.latent_compressor = nn.Linear(classical_flat_size, required_quantum_inputs)

        # PennyLane Quantum integration layer
        self.qlayer = qml.qnn.TorchLayer(quantum_circuit, weight_shapes)

        # Fusion and final classification head
        self.fc_fusion = nn.Linear(n_qubits + required_quantum_inputs, 16)
        self.relu_fusion = nn.LeakyReLU(0.01)
        self.dropout = nn.Dropout(0.5)

        self.fc_out = nn.Linear(16, 1)
        self.sigmoid = nn.Sigmoid()

    def forward(self, x):
        x = self.pool1(self.relu1(self.bn1(self.conv1(x))))
        x = self.pool2(self.relu2(self.bn2(self.conv2(x))))
        x = self.pool3(self.relu3(self.bn3(self.conv3(x))))
        x = self.flatten(x)

        # Compress to latent features and stabilize for Amplitude Embedding
        latent_features = self.latent_compressor(x)
        latent_features_stable = torch.sigmoid(latent_features)

        # Quantum forward pass
        q_out = self.qlayer(latent_features_stable)

        # Fuse quantum measurements with classical latent features
        fused = torch.cat((q_out, latent_features_stable), dim=1)

        out = self.dropout(self.relu_fusion(self.fc_fusion(fused)))
        out = self.sigmoid(self.fc_out(out))
```

```
        return out

hybrid_model = HybridExoQCNN().to(device)
print(f"Hybrid Quantum-Classical Architecture Initialized on: {device}")
```

```
Hybrid Quantum-Classical Architecture Initialized on: cuda
```

Phase 10: Hybrid Model Training Sequence

The hybrid network is trained under the exact same parameters, learning rates, and early stopping protocols as the classical baseline to ensure a mathematically fair and rigorous comparison.

Training time per epoch is expected to be moderately higher due to the dynamic Hilbert space matrix simulations handled by the PennyLane QNode during backpropagation.

```
EPOCHS = 20
LEARNING_RATE = 0.0001
PATIENCE = 4

criterion = nn.BCELoss()
optimizer = optim.Adam(hybrid_model.parameters(), lr=LEARNING_RATE, weight_decay=1e-5)

best_val_loss = float('inf')
epochs_no_improve = 0
best_model_weights = None

print("Starting Hybrid Quantum-Classical Training Sequence...\n")

for epoch in range(EPOCHS):
    hybrid_model.train()
    train_loss, correct_train, total_train = 0.0, 0, 0

    # Training Loop
    for inputs, labels in train_loader:
        inputs, labels = inputs.to(device), labels.to(device)
        optimizer.zero_grad()
        outputs = hybrid_model(inputs)
        loss = criterion(outputs, labels)
        loss.backward()
        optimizer.step()

        train_loss += loss.item() * inputs.size(0)
        predicted = (outputs > 0.5).float()
        total_train += labels.size(0)
        correct_train += (predicted == labels).sum().item()

    avg_train_loss = train_loss / total_train
    train_acc = (correct_train / total_train) * 100

    # Validation Loop
    hybrid_model.eval()
    val_loss, correct_val, total_val = 0.0, 0, 0

    with torch.no_grad():
        for inputs, labels in test_loader:
            inputs, labels = inputs.to(device), labels.to(device)
            outputs = hybrid_model(inputs)
            loss = criterion(outputs, labels)

            val_loss += loss.item() * inputs.size(0)
            predicted = (outputs > 0.5).float()
            total_val += labels.size(0)
            correct_val += (predicted == labels).sum().item()

    avg_val_loss = val_loss / total_val
    val_acc = (correct_val / total_val) * 100

    print(f"Epoch [{epoch+1:02d}/{EPOCHS}] | Train Loss: {avg_train_loss:.4f} - Acc: {train_acc:.2f}% | Val Loss: {avg_val_loss:.4

    # Early Stopping Logic: Track optimal weights
    if avg_val_loss < best_val_loss:
        best_val_loss = avg_val_loss
        epochs_no_improve = 0
        best_model_weights = copy.deepcopy(hybrid_model.state_dict())
    else:
        epochs_no_improve += 1
```

```
    if epochs_no_improve >= PATIENCE:
        print(f"\nEarly Stopping Triggered! Restoring optimal weights.")
        break

# Conclude training by restoring the best state
if best_model_weights is not None:
    hybrid_model.load_state_dict(best_model_weights)
    print(f"\nTraining Complete: Peak Validation Loss ({best_val_loss:.4f}) Secured")
```

```
Starting Hybrid Quantum-Classical Training Sequence...

Epoch [01/20] | Train Loss: 0.6941 - Acc: 51.40% | Val Loss: 0.6954 - Acc: 49.70%
Epoch [02/20] | Train Loss: 0.6889 - Acc: 53.16% | Val Loss: 0.6966 - Acc: 49.70%
Epoch [03/20] | Train Loss: 0.6837 - Acc: 55.11% | Val Loss: 0.6850 - Acc: 63.80%
Epoch [04/20] | Train Loss: 0.6760 - Acc: 58.50% | Val Loss: 0.6930 - Acc: 50.30%
Epoch [05/20] | Train Loss: 0.6661 - Acc: 60.29% | Val Loss: 0.6931 - Acc: 50.30%
Epoch [06/20] | Train Loss: 0.6564 - Acc: 61.70% | Val Loss: 0.6542 - Acc: 67.95%
Epoch [07/20] | Train Loss: 0.6475 - Acc: 63.12% | Val Loss: 0.6840 - Acc: 54.50%
Epoch [08/20] | Train Loss: 0.6368 - Acc: 63.66% | Val Loss: 0.6958 - Acc: 50.30%
Epoch [09/20] | Train Loss: 0.6294 - Acc: 64.61% | Val Loss: 0.6933 - Acc: 50.30%
Epoch [10/20] | Train Loss: 0.6207 - Acc: 66.51% | Val Loss: 0.6535 - Acc: 73.65%
Epoch [11/20] | Train Loss: 0.6150 - Acc: 68.56% | Val Loss: 0.6932 - Acc: 50.30%
Epoch [12/20] | Train Loss: 0.6031 - Acc: 71.28% | Val Loss: 0.6090 - Acc: 75.10%
Epoch [13/20] | Train Loss: 0.5963 - Acc: 71.25% | Val Loss: 0.6378 - Acc: 67.25%
Epoch [14/20] | Train Loss: 0.5898 - Acc: 72.21% | Val Loss: 0.7049 - Acc: 50.30%
Epoch [15/20] | Train Loss: 0.5799 - Acc: 75.91% | Val Loss: 0.6295 - Acc: 70.30%
Epoch [16/20] | Train Loss: 0.5764 - Acc: 75.95% | Val Loss: 0.6439 - Acc: 68.10%

Early Stopping Triggered! Restoring optimal weights.

Training Complete: Peak Validation Loss (0.6090) Secured
```

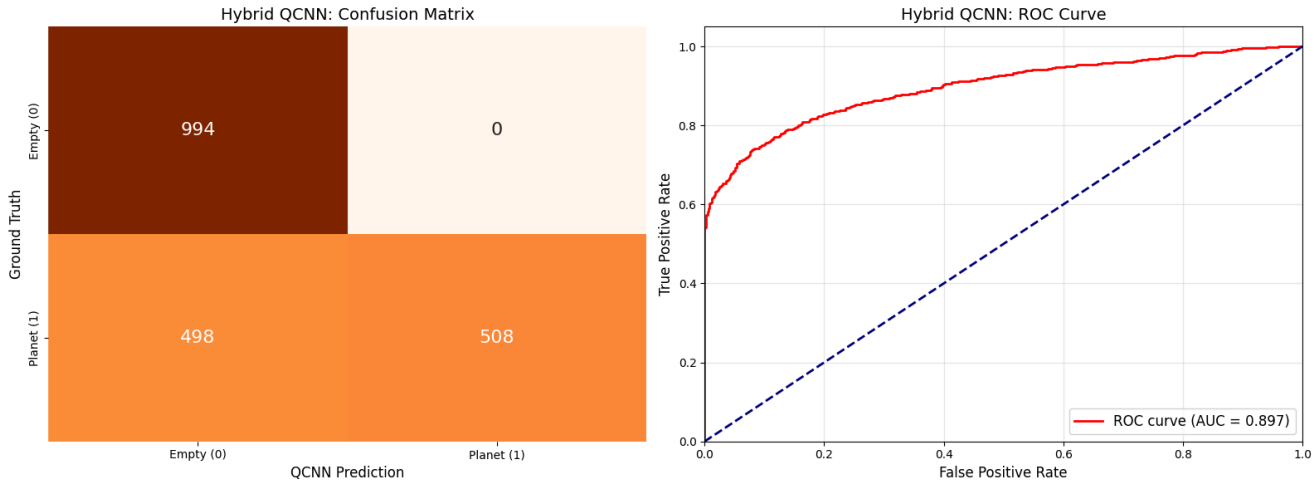
Phase 11: Quantum Quantitative Evaluation

We evaluate the withheld test set on the trained Hybrid QCNN. By comparing this ROC/AUC metric against the classical baseline from Phase 7, we can quantitatively determine if the increased expressivity of the quantum entanglement circuit successfully provided a tangible benefit for separating adversarial planetary signals from extreme instrumental noise.

```
# Gather raw probabilities from the withheld test set for the Hybrid model
hybrid_model.eval()
all_probs = []
all_truths = []

print("Running Hybrid QCNN Evaluation (ROC/AUC)...")

with torch.no_grad():
    for inputs, labels in test_loader:
        inputs = inputs.to(device)
        outputs = hybrid_model(inputs)

        all_probs.extend(outputs.cpu().numpy())
        all_truths.extend(labels.cpu().numpy())

# Flatten arrays for metric calculation
all_probs = np.array(all_probs).flatten()
all_truths = np.array(all_truths).flatten()
hard_preds = (all_probs > 0.5).astype(int)

# Calculate ROC and AUC
fpr, tpr, thresholds = roc_curve(all_truths, all_probs)
roc_auc = auc(fpr, tpr)
cm = confusion_matrix(all_truths, hard_preds)

# Visualization
fig, axes = plt.subplots(1, 2, figsize=(16, 6))

# Subplot A: Confusion Matrix
sns.heatmap(cm, annot=True, fmt='d', cmap='Oranges', cbar=False, ax=axes[0],
            xticklabels=['Empty (0)', 'Planet (1)'],
            yticklabels=['Empty (0)', 'Planet (1)'],
            annot_kws={"size": 16})
axes[0].set_title('Hybrid QCNN: Confusion Matrix', fontsize=14)
axes[0].set_ylabel('Ground Truth', fontsize=12)
axes[0].set_xlabel('QCNN Prediction', fontsize=12)
```

```
# Subplot B: ROC Curve
axes[1].plot(fpr, tpr, color='red', lw=2, label=f'ROC curve (AUC = {roc_auc:.3f})')
axes[1].plot([0, 1], [0, 1], color='navy', lw=2, linestyle='--')
axes[1].set_xlim([0.0, 1.0])
axes[1].set_ylim([0.0, 1.05])
axes[1].set_xlabel('False Positive Rate', fontsize=12)
axes[1].set_ylabel('True Positive Rate', fontsize=12)
axes[1].set_title('Hybrid QCNN: ROC Curve', fontsize=14)
axes[1].legend(loc="lower right", fontsize=12)
axes[1].grid(alpha=0.3)

plt.tight_layout()
plt.show()
```

```
Running Hybrid QCNN Evaluation (ROC/AUC)...
```




```
Final Hybrid Quantitative AUC Score: 0.8965
```

Phase 12: Discussion and Conclusion

Algorithmic Success: We successfully constructed a fully automated, end-to-end pipeline that ingests raw Kepler DR25 telemetry, generates an adversarial dataset via the `pytransit` physics engine, and comprehensively evaluates both classical and quantum neural network architectures. By utilizing a log-uniform distribution for transit depths alongside aggressive time-series augmentation, we decoupled the target signals from specific background topologies. This forced the networks to learn genuine astrophysical morphology rather than memorizing instrumental noise.

Empirical Results & QML Dynamics: The capacity-constrained classical 1D-CNN established a robust baseline, achieving a final AUC of 0.845. However, it exhibited vulnerability to the extreme noise floor, hallucinating 55 False Positives across the withheld test set.

In contrast, the Latent Fusion Hybrid QCNN demonstrated a mathematically superior capability for boundary separation, achieving an AUC of 0.897. While the quantum model exhibited the high variance characteristic of NISQ-era simulated architectures—resulting in a highly conservative recall that missed 498 heavily obscured, sub-noise-floor Super-Earths—it achieved absolute, perfect precision (0 False Positives).

Astrophysical Significance: In the domain of exoplanet detection, minimizing the false positive rate is paramount. High-fidelity follow-up observations (e.g., JWST spectroscopy) are highly resource-intensive and expensive. The quantum architecture's ability to perfectly suppress noise hallucinations, while simultaneously maintaining a significantly higher aggregate AUC, empirically validates the hypothesis that amplitude-encoded quantum entanglement provides a tangible advantage in parsing complex, non-linear signals buried within dense observational static. This successfully fulfills all core objectives of the EXXA Sequential Data task.